House Price Prediction

Findings Summary Report

Overview

This report summarises the analysis of a real-world housing dataset (545 records, 13 columns) sourced from Kaggle. The objective was to build and evaluate regression models that predict residential property prices based on structural and locational features, and to identify the attributes that most strongly drive market value.

Data Preparation

The dataset was found to be clean — no missing values or duplicate records were present. Seven categorical columns (mainroad, guestroom, basement, hotwaterheating, airconditioning, prefarea, and furnishingstatus) were converted into numeric form using one-hot encoding, expanding the feature space to 18 predictors. The data was then split 80/20 into training and test sets before modelling.

Model Performance

Two regression models were trained and evaluated on identical test data:

- Linear Regression - MAE: ₹9,70,043 | RMSE: ₹13,24,507 | R^2 Score: 0.6529
- Random Forest Regressor —-MAE: ₹10,21,546 | RMSE: ₹14,00,566 | R^2 Score: 0.6119

Notably, the simpler Linear Regression model marginally outperformed the Random Forest on all three metrics, achieving an R^2 of 0.65. This means the model successfully explains approximately 65% of the variance in house prices.

Key Findings — Features That Drive Price

The correlation heatmap and model coefficients consistently identified three features as the strongest predictors of house price:

- Total Area (sq. ft.) — the single most influential factor; larger homes command proportionally higher prices.
- Number of Bathrooms — a strong positive predictor, reflecting both space and quality of the property.
- Air Conditioning — binary presence/absence had a surprisingly large impact on price, suggesting buyers place a high premium on this amenity.

Parking spaces and the number of storeys also showed moderate positive correlations, while hot water heating had a near-negligible effect.

Surprising findings

The most unexpected finding was that Linear Regression outperformed the Random Forest - an ensemble method typically more powerful on tabular data. This likely reflects the relatively small dataset size (545 rows) and the near-linear relationships between the key features and price, which the simpler model captures efficiently. Additionally, hot water heating — intuitively a desirable feature — showed little correlation with price, suggesting buyers in this market do not price it as a significant luxury.

Business Recommendation

For a real estate business, the analysis points to two high-impact strategies. First, prioritise acquiring or developing properties with larger floor areas and multiple bathrooms, as these attributes consistently yield higher valuations. Second, consider retrofitting air conditioning in existing mid-range listings as a cost-effective intervention to lift market value — the data suggests buyers are willing to pay a premium for it. These actionable levers can guide both acquisition decisions and value-add renovation planning.

Analysis performed using Python · Scikit-learn · Seaborn · Matplotlib